Technical Report for Layer-Importance-Aware AI Model Downloading in Wireless Networks

A Proof of Theorem 1

Under Assumption 1, the locally smoothed performance response can be expanded around f_0 as

$$\mathbb{E}[\Delta A] = \nabla A(f_0)^T \mathbb{E}[\Delta f] + \frac{1}{2}\mathbb{E}\big[\Delta f^T H \Delta f\big] + o\big(\mathbb{E}\big[\|\Delta f\|^2\big]\big), \quad (43)$$

where

$$\Delta A \triangleq A(\hat{f}) - A(f_0), \quad (44)$$

$\Delta f \triangleq \hat{f} - f_0$, and H denotes the Hessian of $A(\cdot)$ with respect to the logits, evaluated at f_0. This expression provides the link between the logit perturbation and the resulting performance variation, and will be combined with the second-order expansion of $\hat{f}$ with respect to the layer-wise parameter perturbations.

Define the first- and second-order logit perturbation terms as

$$\Delta f^{(1)} \triangleq \sum_{i=1}^{D} J_i \Delta w_i, \qquad \Delta f^{(2)} \triangleq \frac{1}{2}\sum_{i=1}^{D}\sum_{j=1}^{D} \Delta w_i^T G_{i,j} \Delta w_j, \quad (45)$$

such that

$$\hat{f} = f_0 + \Delta f^{(1)} + \Delta f^{(2)} + o(\|\Delta w\|^2). \quad (46)$$

Substituting the above expression into $E[\Delta A]$, and retaining only terms up to second order with respect to Δw, yields

$$\begin{aligned} E[\Delta A] =& \nabla A(f_0)^T E[\Delta f^{(1)}] + \nabla A(f_0)^T E[\Delta f^{(2)}] \\ &+ \frac{1}{2} E\big[(\Delta f^{(1)})^T H \Delta f^{(1)}\big] + o(\|\Delta w\|^2). \end{aligned} \quad (47)$$

Here, the terms involving $(\Delta f^{(2)})^T H \Delta f^{(2)}$ and the cross products between $\Delta f^{(1)}$ and $\Delta f^{(2)}$ are of order higher than two and are therefore omitted.

We first evaluate the linear term. By Assumption 3, $E[\Delta w_i] = 0$ for all i, and thus

$$E[\Delta f^{(1)}] = \sum_{i=1}^{D} J_i E[\Delta w_i] = 0. \quad (48)$$

Therefore, the first term in (47) vanishes.

Next, consider $E[\Delta f^{(2)}]$. For the k-th logit component,

$$[\Delta f^{(2)}]_k = \frac{1}{2}\sum_{i=1}^{D}\sum_{j=1}^{D} \Delta w_i^T G_{i,j}^{(k)} \Delta w_j, \quad (49)$$

where $G_{i,j}^{(k)}$ denotes the (i,j)-th second-order derivative block of the k-th logit component. Taking expectation gives

$$E\big[[\Delta f^{(2)}]_k\big] = \frac{1}{2}\sum_{i=1}^{D}\sum_{j=1}^{D} E\Big[\Delta w_i^T G_{i,j}^{(k)} \Delta w_j\Big]. \quad (50)$$

For $i \neq j$, Assumption 3 implies that Δw_i and Δw_j are independent and zero-mean, hence

$$E\Big[\Delta w_i^T G_{i,j}^{(k)} \Delta w_j\Big] = E[\Delta w_i]^T G_{i,j}^{(k)} E[\Delta w_j] = 0. \quad (51)$$

Therefore, only the self-interaction terms remain:

$$E\big[[\Delta f^{(2)}]_k\big] = \frac{1}{2}\sum_{i=1}^{D} E\Big[\Delta w_i^T G_{i,i}^{(k)} \Delta w_i\Big]. \quad (52)$$

It follows that

$$\nabla A(f_0)^T E[\Delta f^{(2)}] = \frac{1}{2}\sum_{k=1}^{F} \nabla A_{(k)} \sum_{i=1}^{D} E\Big[\Delta w_i^T G_{i,i}^{(k)} \Delta w_i\Big]. \quad (53)$$

Now consider the quadratic term in (47):

$$\begin{aligned} E\big[(\Delta f^{(1)})^T H \Delta f^{(1)}\big] &= E\left[\left(\sum_{i=1}^{D} J_i \Delta w_i\right)^T H \left(\sum_{j=1}^{D} J_j \Delta w_j\right)\right] \\ &= \sum_{i=1}^{D}\sum_{j=1}^{D} E\big[\Delta w_i^T J_i^T H J_j \Delta w_j\big]. \end{aligned} \quad (54)$$

Again, for $i \neq j$, independence and zero-mean imply

$$E\big[\Delta w_i^T J_i^T H J_j \Delta w_j\big] = E[\Delta w_i]^T J_i^T H J_j E[\Delta w_j] = 0. \quad (55)$$

Thus,

$$E\big[(\Delta f^{(1)})^T H \Delta f^{(1)}\big] = \sum_{i=1}^{D} E\big[\Delta w_i^T J_i^T H J_i \Delta w_i\big]. \quad (56)$$

Substituting (53) and (56) into (47), we obtain

$$\begin{aligned} E[\Delta A] &= \frac{1}{2}\sum_{k=1}^{F} \nabla A_{(k)} \sum_{i=1}^{D} E\Big[\Delta w_i^T G_{i,i}^{(k)} \Delta w_i\Big] \\ &\quad + \frac{1}{2}\sum_{i=1}^{D} E\big[\Delta w_i^T J_i^T H J_i \Delta w_i\big] + o(\|\Delta w\|^2) \\ &= \frac{1}{2}\sum_{i=1}^{D} E\left[\Delta w_i^T \left(\sum_{k=1}^{F} \nabla A_{(k)} G_{i,i}^{(k)} + J_i^T H J_i\right) \Delta w_i\right] + o(\|\Delta w\|^2). \end{aligned} \quad (57)$$

Using the definition of M_i in (10), we have

$$E[\Delta A] = \frac{1}{2}\sum_{i=1}^{D} E\big[\Delta w_i^T (M_i + J_i^T H J_i) \Delta w_i\big] + o(\|\Delta w\|^2). \quad (58)$$

Finally, applying the identity

$$E[x^T A x] = \mathrm{tr}(A\, E[x x^T]), \quad (59)$$

for any random vector x and constant matrix A of compatible dimensions, together with $\Sigma_i = E[\Delta w_i \Delta w_i^T]$, yields

$$E[\Delta A] = \frac{1}{2}\sum_{i=1}^{D} \mathrm{tr}\Big((M_i + J_i^T H J_i)\Sigma_i\Big) + o(\|\Delta w\|^2). \quad (60)$$

Neglecting the higher-order remainder term under the small-perturbation assumption completes the proof.

B Proof of Proposition 1

Consider a parameter w transmitted through analog modulation. Let the received signal after retransmission and combining be written as

$$y = w + n_{\mathrm{eq}}, \quad (61)$$

where n_{eq} denotes the equivalent additive noise after combining. Then the reconstruction error is

$$e_w = \hat{w} - w = n_{\text{eq}}. \tag{62}$$

For a given realization of the effective combined SNR γ^{tot}, the conditional distortion power satisfies

$$\mathbb{E}\left[e_w^2 \mid \gamma^{\text{tot}}\right] = \frac{\mathbb{E}[w^2]}{\gamma^{\text{tot}}}, \tag{63}$$

since the received SNR is defined as the ratio between the signal power and the equivalent noise power after combining. Taking expectation over γ^{tot}, we have

$$\begin{aligned}\mathbb{E}[e_w^2] &= \mathbb{E}_{\gamma^{\text{tot}}}\left[\mathbb{E}\left[e_w^2 \mid \gamma^{\text{tot}}\right]\right] \\ &= \mathbb{E}_{\gamma^{\text{tot}}}\left[\frac{\mathbb{E}[w^2]}{\gamma^{\text{tot}}}\right] \\ &= \mathbb{E}[w^2]\,\mathbb{E}\left[\frac{1}{\gamma^{\text{tot}}}\right].\end{aligned} \tag{64}$$

Applying the same relation to all parameters in layer (i, j), the normalized layer distortion becomes

$$\begin{aligned}\frac{1}{LSNR_{i,j}} &= \frac{\mathbb{E}\left[\|\Delta \mathbf{w}_{i,j}\|^2\right]}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]} \\ &= \mathbb{E}\left[\frac{1}{\gamma_{i,j}^{\text{tot}}}\right].\end{aligned} \tag{65}$$

This completes the proof.

C Proof of Proposition 2

For the j-th layer of the i-th UE, by the LSNR definition, we have

$$\frac{1}{LSNR_{i,j}} = \frac{\mathbb{E}\left[\|\Delta \mathbf{w}_{i,j}\|^2\right]}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]}. \tag{66}$$

Furthermore, define $s_k \in \{0, 1\}$ as the error indicator of the k-th bit, where $s_k = 1$ indicates that the bit is flipped during transmission and $s_k = 0$ otherwise. Then the reconstruction error of this parameter can be expressed as

$$e_w = \hat{w} - w = \Delta \sum_{k=0}^{b-1} s_k (1 - 2n_k) 2^k. \tag{67}$$

Substituting (67) into the numerator gives

$$\begin{aligned}\mathbb{E}\left[\|\Delta \mathbf{w}_{i,j}\|^2\right] &= \mathbb{E}\left[\sum_{\ell=1}^{N_{i,j}} \left(\Delta \sum_{k=0}^{b-1} s_{\ell,k}(1 - 2n_{\ell,k}) 2^k\right)^2\right] \\ &= \Delta^2 \mathbb{E}\left[\sum_{\ell=1}^{N_{i,j}} \left(\sum_{k=0}^{b-1} s_{\ell,k}(1 - 2n_{\ell,k}) 2^k\right)^2\right],\end{aligned} \tag{68}$$

where $n_{\ell,k}$ and $s_{\ell,k}$ denote the original bit and the error indicator of the k-th bit of the ℓ-th parameter, respectively.

Expanding the square term yields

$$\begin{aligned}\left(\sum_{k=0}^{b-1} s_{\ell,k}(1 - 2n_{\ell,k}) 2^k\right)^2 &= \sum_{k=0}^{b-1} s_{\ell,k}^2 (1 - 2n_{\ell,k})^2 4^k \\ &\quad + \sum_{k \neq t} s_{\ell,k} s_{\ell,t} (1 - 2n_{\ell,k})(1 - 2n_{\ell,t}) 2^{k+t}.\end{aligned} \tag{69}$$

Since $s_{\ell,k} \in \{0, 1\}$, we have $s_{\ell,k}^2 = s_{\ell,k}$, and since $n_{\ell,k} \in \{0, 1\}$, we have $(1 - 2n_{\ell,k})^2 = 1$. Therefore, the first term becomes

$$\sum_{k=0}^{b-1} s_{\ell,k} 4^k. \tag{70}$$

Under the assumption that bit flips are independent and identically distributed with probability $P_{b,i,j}$, we have

$$\mathbb{E}[s_{\ell,k}] = P_{b,i,j}, \tag{71}$$

and thus

$$\mathbb{E}\left[\sum_{k=0}^{b-1} s_{\ell,k} 4^k\right] = P_{b,i,j} \sum_{k=0}^{b-1} 4^k. \tag{72}$$

For the cross terms with $k \neq t$, their expectation is proportional to

$$\mathbb{E}[s_{\ell,k} s_{\ell,t}] = P_{b,i,j}^2. \tag{73}$$

Hence, these terms are of order $O(P_{b,i,j}^2)$. In the practical operating regime where $P_{b,i,j} \ll 1$, such higher-order terms are negligible compared with the linear term in $P_{b,i,j}$. Therefore,

$$\mathbb{E}\left[\|\Delta \mathbf{w}_{i,j}\|^2\right] \approx N_{i,j} \Delta^2 P_{b,i,j} \sum_{k=0}^{b-1} 4^k. \tag{74}$$

Substituting (74) into (66), we obtain

$$\begin{aligned}\frac{1}{LSNR_{i,j}} &\approx \frac{N_{i,j} \Delta^2 P_{b,i,j} \sum_{k=0}^{b-1} 4^k}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]} \\ &= \underbrace{\frac{N_{i,j} \Delta^2 \sum_{k=0}^{b-1} 4^k}{\mathbb{E}\left[\|\mathbf{w}_{i,j}\|^2\right]}}_{\zeta_{i,j}} P_{b,i,j}.\end{aligned} \tag{75}$$

This proves (22).

D Empirical Estimation of α_i and β_i

The estimation procedure of α_i and β_i is outlined in Algorithm 1.

E Proof of Theorem 2

Introduce $z_{i,j} = T_{i,j}(\mathbf{B}) P_{i,j}$ (To simplify, we use $T_{i,j}$ instead of $T_{i,j}(\mathbf{B})$ in following proofs), the analog transmission problem can be equivalently rewritten as

$$\min_{\mathbf{z}} \quad \frac{1}{\sum_{i=1}^m r_i} \sum_{i=1}^m \sum_{j=1}^{D_i} \frac{r_i \beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 z_{i,j}} \tag{76}$$

subject to

$$\sum_{i=1}^m \sum_{j=1}^{D_i} c_{i,j} N_{i,j} z_{i,j} \leq \bar{P} C_m, \qquad z_{i,j} > 0, \ \forall i, j, \tag{77}$$

where C_m denotes the total symbol budget, i.e., the maximum number of symbols that can be transmitted when the available bandwidth is fully utilized. Under the real-valued-symbol convention, we have $C_m = 2Bt_m$. Since the bandwidth allocation variables $T_{i,j}$ do not appear explicitly in either the objective function or the power constraint after the substitution, the bandwidth-related constraints in (2) and (3) can be handled separately after the optimal closed-form solution of $\mathbf{z}$ is obtained.

Algorithm 1 Empirical Estimation of α_i and β_i

Require: Model $f(\cdot;\mathbf{w})$, validation set $\mathcal{D}_{\text{val}}$, samples N_{s}, prescribed SNR_i, thresholds $\Delta Acc_{\min}$ and $\Delta Acc_{\max}$
Ensure: $\{\alpha_i, \beta_i\}_{i=1}^{D}$

Compute the original logits $f_0(x)$ and accuracy Acc_0 on $\mathcal{D}_{\text{val}}$
for each layer $i = 1, \ldots, D$ **do**
  Set $S_\alpha = 0$, $S_\beta = 0$, and $C_\beta = 0$
  for $n = 1, \ldots, N_{\text{s}}$ **do**
    Inject random noise into layer i according to SNR_i
    Compute logits $\hat{f}_i^{(n)}(x)$ and accuracy $\widehat{Acc}_i^{(n)}$
    Compute
$$D_i^{(n)} = \frac{1}{|\mathcal{D}_{\text{val}}|} \sum_{x \in \mathcal{D}_{\text{val}}} \left\| \hat{f}_i^{(n)}(x) - f_0(x) \right\|_2^2$$
    Update $S_\alpha \leftarrow S_\alpha + D_i^{(n)} LSNR_i$
    Compute $\Delta Acc_i^{(n)} = Acc_0 - \widehat{Acc}_i^{(n)}$
    if $\Delta Acc_{\min} \le \Delta Acc_i^{(n)} \le \Delta Acc_{\max}$ **then**
      Update $S_\beta \leftarrow S_\beta + \Delta Acc_i^{(n)} / D_i^{(n)}$
      Update $C_\beta \leftarrow C_\beta + 1$
    end if
  end for
  Set $\alpha_i = S_\alpha / N_{\text{s}}$
  Set $\beta_i = S_\beta / C_\beta$
end for
return $\{\alpha_i, \beta_i\}_{i=1}^{D}$

Since $1/z_{i,j}$ is convex over $z_{i,j} > 0$, problem (76)–(77) is convex. Moreover, the objective function is strictly decreasing with respect to each $z_{i,j}$. Therefore, the power constraint in (77) must be active at the optimum, i.e.,

$$\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} z_{i,j} = \bar{P} C_m. \tag{78}$$

For convenience, define

$$a_{i,j} \triangleq \sqrt{\frac{r_i \beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 c_{i,j} N_{i,j}}}. \tag{79}$$

The Lagrangian of the problem is

$$\mathcal{L}(\mathbf{z}, \lambda) = \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \sum_{j=1}^{D_i} \frac{r_i \beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 z_{i,j}} + \lambda \left(\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} z_{i,j} - \bar{P} C_m \right). \tag{80}$$

From the stationarity condition, we have

$$-\frac{r_i \beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 z_{i,j}^2 \sum_{i=1}^{m} r_i} + \lambda c_{i,j} N_{i,j} = 0, \tag{81}$$

which gives

$$z_{i,j} = \frac{a_{i,j}}{\sqrt{\lambda \sum_{i=1}^{m} r_i}}. \tag{82}$$

Substituting (82) into (78) yields

$$\frac{1}{\sqrt{\lambda \sum_{i=1}^{m} r_i}} = \frac{\bar{P} C_m}{\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j}}. \tag{83}$$

Hence, the optimal solution is

$$z_{i,j}^* = \frac{\bar{P} C_m \, a_{i,j}}{\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j}}, \tag{84}$$

which proves (27).

Substituting $z_{i,j}^*$ into the weighted accuracy-loss expression gives

$$\Delta acc_i = \frac{1}{r_i} \sum_{j=1}^{D_i} \frac{\beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2 z_{i,j}^*} \tag{85}$$

$$= \frac{1}{r_i} \sum_{j=1}^{D_i} \frac{\beta_{i,j} \alpha_{i,j} \sigma_i^2}{h_i^2} \cdot \frac{\sum_{u=1}^{m} \sum_{v=1}^{D_u} c_{u,v} N_{u,v} a_{u,v}}{\bar{P} C_m \, a_{i,j}} \tag{86}$$

$$= \frac{\sum_{u=1}^{m} \sum_{v=1}^{D_u} c_{u,v} N_{u,v} a_{u,v}}{\bar{P} C_m \, r_i} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j}, \tag{87}$$

which proves (29).

Therefore, the overall weighted accuracy loss is

$$\Delta acc = \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} r_i \Delta acc_i \tag{88}$$

$$= \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \frac{\sum_{u=1}^{m} \sum_{v=1}^{D_u} c_{u,v} N_{u,v} a_{u,v}}{\bar{P} C_m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j} \tag{89}$$

$$= \frac{\left(\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} a_{i,j} \right)^2}{\bar{P} C_m \sum_{i=1}^{m} r_i}, \tag{90}$$

which proves (30). The proof is thus completed.

F Proof of Theorem 3

Introduce $z_{i,j} = T_{i,j} P_{i,j}$, the digital transmission problem can be equivalently rewritten as

$$\min_{\mathbf{z}} \quad \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \sum_{j=1}^{D_i} d_{i,j} Q\left(\sqrt{e_{i,j} z_{i,j}}\right) \tag{91}$$

subject to

$$\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} z_{i,j} \le \bar{P} C_m, \qquad z_{i,j} > 0, \ \forall i, j, \tag{92}$$

where

$$d_{i,j} = \frac{r_i \beta_{i,j} \alpha_{i,j} \zeta_{i,j} K_{i,j}}{c_{i,j} N_{i,j}}, \qquad e_{i,j} = \frac{\eta_{i,j} h_i^2}{\sigma_i^2}. \tag{93}$$

Since the bandwidth allocation variables $T_{i,j}$ do not appear explicitly in either the objective function or the power constraint after the substitution, the bandwidth-related constraints in (2) and (3) can be handled separately after the optimal semi-closed-form solution of $\mathbf{z}$ is obtained.

To establish convexity, define

$$f_{i,j}(z) = Q\left(\sqrt{e_{i,j} z}\right), \qquad z > 0.$$

Its first derivative is

$$f'_{i,j}(z) = -\frac{\sqrt{e_{i,j}}}{2\sqrt{2\pi z}} \exp\left(-\frac{e_{i,j} z}{2}\right), \tag{94}$$

and its second derivative is

$$f''_{i,j}(z) = \frac{\sqrt{e_{i,j}}}{4\sqrt{2\pi}} \exp\left(-\frac{e_{i,j} z}{2}\right) \left(z^{-3/2} + e_{i,j} z^{-1/2}\right) > 0. \tag{95}$$

Therefore, $f_{i,j}(z)$ is convex for $z > 0$, and hence problem (91)–(92) is convex. Moreover, the objective function is strictly decreasing with respect to each $z_{i,j}$. Therefore, the power constraint in (92) must be active at the optimum, i.e.,

$$\sum_{i=1}^{m}\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j} = \bar{P}\,C_m. \tag{96}$$

The Lagrangian of the problem is

$$\mathcal{L}(\mathbf{z},\lambda) = \frac{1}{\sum_{i=1}^{m} r_i}\sum_{i=1}^{m}\sum_{j=1}^{D_i} d_{i,j}Q\big(\sqrt{e_{i,j}z_{i,j}}\big)+\lambda\left(\sum_{i=1}^{m}\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j} - \bar{P}C_m\right). \tag{97}$$

From the stationarity condition, we have

$$-\frac{d_{i,j}\sqrt{e_{i,j}}}{2\sqrt{2\pi z_{i,j}}\,\sum_{i=1}^{m} r_i}\exp\Big(-\frac{e_{i,j}z_{i,j}}{2}\Big) + \lambda c_{i,j}N_{i,j} = 0. \tag{98}$$

Equivalently,

$$\frac{d_{i,j}\sqrt{e_{i,j}}}{2\sqrt{2\pi z_{i,j}}\,\sum_{i=1}^{m} r_i}\exp\Big(-\frac{e_{i,j}z_{i,j}}{2}\Big) = \lambda c_{i,j}N_{i,j}. \tag{99}$$

Multiplying both sides of (99) by $\sum_{i=1}^{m} r_i$ and absorbing the constant factor into the Lagrange multiplier, (99) can be rewritten as

$$\frac{d_{i,j}\sqrt{e_{i,j}}}{2\sqrt{2\pi z_{i,j}}}\exp\Big(-\frac{e_{i,j}z_{i,j}}{2}\Big) = \lambda c_{i,j}N_{i,j}. \tag{100}$$

Squaring both sides of (100) and rearranging terms yield

$$(e_{i,j}z_{i,j})\exp(e_{i,j}z_{i,j}) = \frac{d_{i,j}^2 e_{i,j}^2}{8\pi\lambda^2 c_{i,j}^2 N_{i,j}^2}. \tag{101}$$

Thus, by the definition of the Lambert-W function,

$$z_{i,j}^* = \frac{1}{e_{i,j}}W\left(\frac{d_{i,j}^2 e_{i,j}^2}{8\pi\lambda^2 c_{i,j}^2 N_{i,j}^2}\right). \tag{102}$$

By further absorbing $c_{i,j}N_{i,j}$ into the multiplier, (102) can be equivalently expressed as

$$z_{i,j}^* = \frac{1}{e_{i,j}}W\left(\frac{d_{i,j}^2 e_{i,j}^2}{8\pi(\lambda^*)^2}\right), \tag{103}$$

which proves (35).

Substituting $z_{i,j}^*$ into the active constraint (96) gives

$$\sum_{i=1}^{m}\sum_{j=1}^{D_i}\frac{c_{i,j}N_{i,j}}{e_{i,j}}W\left(\frac{d_{i,j}^2 e_{i,j}^2}{8\pi(\lambda^*)^2}\right) = \bar{P}\,C_m. \tag{104}$$

The scalar λ^* can be efficiently obtained by solving the resulting one-dimensional monotone equation via bisection. The proof is thus completed.

G Proof of Proposition 3

Under Rayleigh fading, the instantaneous received SNR is exponentially distributed as

$$\gamma \sim \mathrm{Exp}(\bar{\gamma}),$$

where $\bar{\gamma}$ denotes the average received SNR per transmission. After k transmissions with MRC, the combined SNR s_k follows a Gamma distribution with PDF

$$f_{s,k}(s) = \frac{s^{k-1}}{\bar{\gamma}^k(k-1)!}e^{-s/\bar{\gamma}}, \qquad s \geq 0. \tag{105}$$

Let Γ denote the stopping threshold. The event $T = k$ means that the accumulated SNR is below Γ after $k-1$ transmissions and reaches Γ at the k-th transmission. Hence, for $s_0 \geq \Gamma$,

$$\begin{aligned} g_{s,T}(s = s_0, T = k) &= \int_0^{\Gamma} f_{s,k-1}(s)f_{s,1}(s_0 - s)\,ds \\ &= \int_0^{\Gamma}\frac{s^{k-2}}{\bar{\gamma}^{k-1}(k-2)!}e^{-s/\bar{\gamma}}\cdot\frac{1}{\bar{\gamma}}e^{-(s_0-s)/\bar{\gamma}}\,ds \\ &= \frac{\Gamma^{k-1}}{\bar{\gamma}^k(k-1)!}e^{-s_0/\bar{\gamma}}. \end{aligned} \tag{106}$$

Therefore, the PDF of the stopping SNR is

$$\begin{aligned} f_s(s) &= \sum_{k=1}^{\infty} g_{s,T}(s = s, T = k) \\ &= \sum_{k=1}^{\infty}\frac{\Gamma^{k-1}}{\bar{\gamma}^k(k-1)!}e^{-s/\bar{\gamma}} \\ &= \frac{1}{\bar{\gamma}}e^{-s/\bar{\gamma}}\sum_{k=1}^{\infty}\frac{(\Gamma/\bar{\gamma})^{k-1}}{(k-1)!} \\ &= \frac{1}{\bar{\gamma}}e^{-(s-\Gamma)/\bar{\gamma}}, \qquad s \geq \Gamma, \end{aligned} \tag{107}$$

which is a shifted exponential distribution.

The stopping probability at the k-th transmission is

$$\begin{aligned} \Pr(T = k) &= \int_{\Gamma}^{\infty} g_{s,T}(s = s, T = k)\,ds \\ &= \frac{(\Gamma/\bar{\gamma})^{k-1}}{(k-1)!}e^{-\Gamma/\bar{\gamma}}. \end{aligned} \tag{108}$$

Hence, $T - 1 \sim \mathrm{Poisson}(\Gamma/\bar{\gamma})$, and

$$\mathbb{E}[T] = 1 + \frac{\Gamma}{\bar{\gamma}}. \tag{109}$$

For the analog case,

$$\begin{aligned} \frac{1}{LSNR} &= \mathbb{E}\left[\frac{1}{\gamma_{\mathrm{tot}}}\right] \\ &= \int_{\Gamma}^{\infty}\frac{1}{\gamma_{\mathrm{tot}}}\frac{1}{\bar{\gamma}}e^{-(\gamma_{\mathrm{tot}}-\Gamma)/\bar{\gamma}}\,d\gamma_{\mathrm{tot}} \\ &= \frac{1}{\bar{\gamma}}e^{\Gamma/\bar{\gamma}}\int_{\Gamma}^{\infty}\frac{1}{\gamma_{\mathrm{tot}}}e^{-\gamma_{\mathrm{tot}}/\bar{\gamma}}\,d\gamma_{\mathrm{tot}} \\ &= \frac{1}{\bar{\gamma}}e^{\Gamma/\bar{\gamma}}E_1\left(\frac{\Gamma}{\bar{\gamma}}\right), \end{aligned} \tag{110}$$

where $E_1(x) = \int_x^{\infty}\frac{e^{-t}}{t}\,dt$ is the exponential integral.

H Proof of Theorem 4

H.1 Convexity of $\sqrt{G(x)}$

Lemma 1. *For*

$$G(x) = xe^{x-1}E_1(x-1), \qquad x > 1,$$

the function $\sqrt{G(x)}$ *is strictly convex on* $(1,\infty)$.

Proof. Let

$$t = x - 1 > 0, \qquad y(t) = e^t E_1(t).$$

Then

$$G(x) = xe^{x-1}E_1(x-1) = (t+1)y(t),$$

and hence

$$\sqrt{G(x)} = \sqrt{(t+1)y(t)}.$$

Since $dt/dx = 1$, it suffices to prove that $\sqrt{(t+1)y(t)}$ is strictly convex for $t > 0$.

Using

$$y'(t) = y(t) - \frac{1}{t},$$

we obtain

$$\frac{d^2}{dt^2}\sqrt{(t+1)y(t)} = \frac{\Phi(t,y(t))}{4t^2(t+1)^{3/2}y(t)^{3/2}},$$

where

$$\Phi(t,y) = t^2(t^2+4t+2)y^2 + 2(t+1)y - (t+1)^2.$$

Since $t > 0$ and $y(t) > 0$, the denominator is positive. Thus it remains to show

$$\Phi(t,y(t)) > 0.$$

Moreover, $\Phi(t,y)$ is increasing in y, because

$$\frac{\partial\Phi(t,y)}{\partial y} = 2t^2(t^2+4t+2)y + 2(t+1) > 0.$$

From the continued-fraction expansion of $e^tE_1(t)$, we have the lower bound

$$y(t) = e^tE_1(t) \geq \frac{t+3}{t^2+4t+2}, \qquad t > 0.$$

Therefore,

$$\Phi(t,y(t)) \geq \Phi\left(t, \frac{t+3}{t^2+4t+2}\right) = \frac{4}{t^2+4t+2} > 0.$$

Hence

$$\frac{d^2}{dt^2}\sqrt{(t+1)y(t)} > 0, \qquad t > 0,$$

and equivalently

$$\frac{d^2}{dx^2}\sqrt{G(x)} > 0, \qquad x > 1.$$

Thus, $\sqrt{G(x)}$ is strictly convex on $(1,\infty)$. □

H.2 Semi-closed-form for P4

Introduce $z_{i,j} = T_{i,j}P_{i,j}$, then P4 can be equivalently rewritten as

$$\min_{\mathbf{T},\mathbf{z}} \quad \frac{1}{\sum_{i=1}^m r_i}\sum_{i=1}^m\sum_{j=1}^{D_i}\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2}\frac{G(T_{i,j})}{z_{i,j}} \tag{111}$$

subject to

$$\sum_{i=1}^k\sum_{j=1}^{D_i} c_{i,j}N_{i,j}T_{i,j} \leq C_k, \qquad \forall k, \tag{112}$$

$$\sum_{i=1}^m\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j} \leq \bar{P}C_m, \tag{113}$$

where (113) follows from the fact that the last deadline constraint is active at optimum. $C_k = 2Bt_k$ denotes the maximum number of available real-valued symbols before deadline t_k.

For given $\mathbf{T}$, the subproblem with respect to $\mathbf{z}$ is convex. Its Lagrangian is

$$\mathcal{L}_1 = \sum_{i=1}^m\sum_{j=1}^{D_i}\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2}\frac{G(T_{i,j})}{z_{i,j}} + \lambda\left(\sum_{i=1}^m\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j} - \bar{P}C_m\right), \tag{114}$$

where $\lambda \geq 0$. From the KKT stationarity condition,

$$-\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2}\frac{G(T_{i,j})}{z_{i,j}^2} + \lambda c_{i,j}N_{i,j} = 0, \tag{115}$$

which yields

$$z_{i,j}^* = \sqrt{\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2\, G(T_{i,j})}{\lambda h_i^2 c_{i,j}N_{i,j}}}. \tag{116}$$

Since (113) is active at optimum, λ is determined by

$$\sum_{i=1}^m\sum_{j=1}^{D_i} c_{i,j}N_{i,j}z_{i,j}^* = \bar{P}C_m, \tag{117}$$

and thus

$$z_{i,j}^* = \frac{\bar{P}C_m\sqrt{\dfrac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2\, G(T_{i,j})}{h_i^2 c_{i,j}N_{i,j}}}}{\sum\limits_{u=1}^m\sum\limits_{v=1}^{D_u}\sqrt{\dfrac{r_u\beta_{u,v}\alpha_{u,v}\sigma_u^2}{h_u^2}\, c_{u,v}N_{u,v}G(T_{u,v})}}. \tag{118}$$

Define

$$a_{i,j} \triangleq \sqrt{\frac{r_i\beta_{i,j}\alpha_{i,j}\sigma_i^2}{h_i^2 c_{i,j}N_{i,j}}}. \tag{119}$$

Then (118) can be rewritten as

$$z_{i,j}^* = \frac{\bar{P}C_m\, a_{i,j}\sqrt{G(T_{i,j})}}{\sum\limits_{u=1}^m\sum\limits_{v=1}^{D_u} a_{u,v}c_{u,v}N_{u,v}\sqrt{G(T_{u,v})}}. \tag{120}$$

Substituting (120) into (111), the original problem reduces to

$$\min_{\mathbf{T}} \quad \sum_{i=1}^m\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\phi(T_{i,j}) \tag{121}$$

subject to (112), where

$$\phi(T) \triangleq \sqrt{G(T)}. \tag{122}$$

Since $\phi(T) = \sqrt{G(T)}$ is convex, thus (121) is a convex optimization problem.

Let $\mu_k \geq 0$ denote the dual variable associated with the k-th deadline constraint in (112). The corresponding Lagrangian is

$$\mathcal{L}_2 = \sum_{i=1}^m\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\phi(T_{i,j}) + \sum_{k=1}^m\mu_k\left(\sum_{i=1}^k\sum_{j=1}^{D_i} c_{i,j}N_{i,j}T_{i,j} - C_k\right). \tag{123}$$

Since $T_{i,j}$ appears in all cumulative constraints with indices $k \geq i$, the stationarity condition gives

$$a_{i,j}\phi'(T_{i,j}^*) + \sum_{k\geq i}\mu_k = 0. \tag{124}$$

Define

$$\Lambda_i \triangleq \sum_{k\geq i} \mu_k, \qquad \psi(T) \triangleq -\phi'(T) = -\frac{G'(T)}{2\sqrt{G(T)}}. \tag{125}$$

Then

$$\psi(T^*_{i,j}) = \frac{\Lambda_i}{a_{i,j}}, \tag{126}$$

which leads to

$$T^*_{i,j} = \psi^{-1}\left(\frac{\Lambda_i}{a_{i,j}}\right). \tag{127}$$

Together with (120), this proves the semi-closed-form solution.

Finally, substituting (127) into the reduced objective yields

$$\Delta acc = \frac{\left(\sum_{i=1}^{m}\sum_{j=1}^{D_i} a_{i,j}c_{i,j}N_{i,j}\sqrt{G(T^*_{i,j})}\right)^2}{\bar{P}C_m \sum_{i=1}^{m} r_i}. \tag{128}$$

The semi-closed-form solution can be efficiently computed via an active-set search over the cumulative deadline constraints. Specifically, starting from the last deadline constraint, the active set is progressively updated by adding violated constraints and removing those with negative multipliers, until the KKT conditions are satisfied. The proof is completed.

I Proof of Proposition 4

For the digital transmission scheme, the instantaneous BER is approximated by

$$\hat{P}_b = KQ\left(\sqrt{\eta\gamma_{\text{tot}}}\right), \tag{129}$$

where K and η are modulation-related constants. Under the threshold-based SNR control strategy, the stopping SNR γ_{tot} follows the shifted exponential distribution

$$f_{\gamma_{\text{tot}}}(\gamma) = \frac{1}{\bar{\gamma}}\exp\left(-\frac{\gamma-\Gamma}{\bar{\gamma}}\right), \qquad \gamma \geq \Gamma, \tag{130}$$

where Γ is the stopping threshold and $\bar{\gamma}$ is the average received SNR of each transmission round. Therefore, the average BER is given by

$$\begin{aligned} P_b &= \int_{\Gamma}^{\infty} \hat{P}_b \frac{1}{\bar{\gamma}} e^{-\frac{\gamma_{\text{tot}}-\Gamma}{\bar{\gamma}}}\, d\gamma_{\text{tot}} \\ &= \int_{\Gamma}^{\infty} KQ\left(\sqrt{\eta\gamma_{\text{tot}}}\right) \frac{1}{\bar{\gamma}} e^{-\frac{\gamma_{\text{tot}}-\Gamma}{\bar{\gamma}}}\, d\gamma_{\text{tot}}. \end{aligned} \tag{131}$$

By Craig's formula,

$$Q(x) = \frac{1}{\pi}\int_0^{\pi/2} \exp\left(-\frac{x^2}{2\sin^2\theta}\right) d\theta, \tag{132}$$

we have

$$\begin{aligned} P_b &= \frac{K}{\pi\bar{\gamma}} e^{\frac{\Gamma}{\bar{\gamma}}} \int_{\Gamma}^{\infty}\int_0^{\pi/2} \exp\left(-\frac{\eta\gamma_{\text{tot}}}{2\sin^2\theta} - \frac{\gamma_{\text{tot}}}{\bar{\gamma}}\right) d\theta\, d\gamma_{\text{tot}} \\ &= \frac{K}{\pi\bar{\gamma}} e^{\frac{\Gamma}{\bar{\gamma}}} \int_0^{\pi/2}\int_{\Gamma}^{\infty} \exp\left(-\left(\frac{\eta}{2\sin^2\theta} + \frac{1}{\bar{\gamma}}\right)\gamma_{\text{tot}}\right) d\gamma_{\text{tot}}\, d\theta \\ &= \frac{K}{\pi}\int_0^{\pi/2} \frac{\exp\left(-\frac{\eta\Gamma}{2\sin^2\theta}\right)}{1+\frac{\eta\bar{\gamma}}{2\sin^2\theta}} d\theta. \end{aligned} \tag{133}$$

Next, define

$$b = \frac{\eta\Gamma}{2}, \qquad c = \frac{\eta\bar{\gamma}}{2}. \tag{134}$$

Then

$$\begin{aligned} P_b &= \frac{K}{\pi}\int_0^{\pi/2} \frac{\sin^2\theta}{\sin^2\theta + c} \exp\left(-\frac{b}{\sin^2\theta}\right) d\theta \\ &= \frac{K}{\pi}\int_0^{\pi/2} \exp\left(-\frac{b}{\sin^2\theta}\right) d\theta - \frac{Kc}{\pi}\int_0^{\pi/2} \frac{\exp\left(-\frac{b}{\sin^2\theta}\right)}{\sin^2\theta + c} d\theta. \end{aligned} \tag{135}$$

Using the integral identities

$$\frac{1}{\pi}\int_0^{\pi/2} \exp\left(-\frac{b}{\sin^2\theta}\right) d\theta = Q\left(\sqrt{2b}\right), \tag{136}$$

and

$$\frac{c}{\pi}\int_0^{\pi/2} \frac{\exp\left(-\frac{b}{\sin^2\theta}\right)}{\sin^2\theta + c} d\theta = \sqrt{\frac{c}{c+1}}\, e^{\frac{b}{c}} Q\left(\sqrt{2b\left(1+\frac{1}{c}\right)}\right), \tag{137}$$

it follows that

$$P_b = K\left(Q\left(\sqrt{2b}\right) - \sqrt{\frac{c}{c+1}}\, e^{\frac{b}{c}} Q\left(\sqrt{2b\left(1+\frac{1}{c}\right)}\right)\right). \tag{138}$$

Substituting $b = \eta\Gamma/2$ and $c = \eta\bar{\gamma}/2$ yields

$$P_b = K\left(Q\left(\sqrt{\eta\Gamma}\right) - \sqrt{\frac{\eta\bar{\gamma}}{\eta\bar{\gamma}+2}}\, e^{\frac{\Gamma}{\bar{\gamma}}} Q\left(\sqrt{(\eta\bar{\gamma}+2)\frac{\Gamma}{\bar{\gamma}}}\right)\right). \tag{139}$$

This completes the proof.

J AO-SCA Algorithm for (P5)

To facilitate the algorithm design, we rewrite the bandwidth allocation variables into the equivalent number of transmissions. Therefore, optimizing the bandwidth allocation $\{B_{i,j,k}\}$ is equivalent to optimizing $\{T_{i,j}\}$ under the corresponding deadline-induced symbol constraints. In particular, the cumulative deadline constraints can be written as

$$\sum_{i=1}^{k}\sum_{j=1}^{D_i} c_{i,j}N_{i,j}T_{i,j} \leq C_k, \quad k = 1,\ldots,m, \tag{140}$$

where $C_k = 2Bt_k$ denotes the maximum number of available symbols before deadline t_k.

For the digital transmissions under Rayleigh channels, the average received SNR and the stopping threshold can be rewritten as

$$\bar{\gamma}_{i,j} = \rho_i P_{i,j}, \qquad \Gamma_{i,j} = (T_{i,j}-1)\rho_i P_{i,j}, \tag{141}$$

where $\rho_i = \sigma^2_{h_i}/\sigma^2_i$. Substituting (141) into the expected BER expression yields

$$\begin{aligned} P_{b,i,j}(T_{i,j}, P_{i,j}) = K_{i,j}\Bigg[&Q\left(\sqrt{\eta_{i,j}\rho_i P_{i,j}(T_{i,j}-1)}\right) \\ &- \sqrt{\frac{\eta_{i,j}\rho_i P_{i,j}}{\eta_{i,j}\rho_i P_{i,j}+2}}\, e^{T_{i,j}-1} Q\left(\sqrt{(\eta_{i,j}\rho_i P_{i,j}+2)(T_{i,j}-1)}\right)\Bigg]. \end{aligned} \tag{142}$$

The resulting problem is still difficult to solve directly because $T_{i,j}$ and $P_{i,j}$ are coupled in the expected BER expression. We therefore adopt an alternating optimization framework. At the n-th outer iteration, the transmission-time variables are first optimized with the power allocation fixed at $\mathbf{P}^{(n)}$, and then the power allocation

is optimized with the updated transmission-time variables fixed at $\mathbf{T}^{(n+1)}$. Given the power allocation $\mathbf{P}^{(n)}$, the transmission-time subproblem is defined as

$$\text{(P5-T)} \quad \min_{\mathbf{T}} \quad \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \sum_{j=1}^{D_i} r_i \alpha_{i,j} \beta_{i,j} \zeta_{i,j} P_{b,i,j}(T_{i,j}, P_{i,j}^{(n)}) \quad (143a)$$

$$\text{s.t.} \quad \sum_{i=1}^{k} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} T_{i,j} \le C_k, \quad k = 1, \ldots, m, \quad (143b)$$

$$\sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} \left(P_{i,j}^{(n)} - \bar{P} \right) T_{i,j} \le 0, \quad (143c)$$

$$T_{i,j} \ge 1, \quad \forall i, j. \quad (143d)$$

The objective of P5-T is still nonconvex due to the complicated dependence of the expected BER on $T_{i,j}$. To simplify notation, we omit the indices (i, j) and write the BER term as $P_b(T, P)$. With P fixed, the first-order approximation of $P_b(T, P)$ at the current SCA point $T^{(\ell)}$ is

$$\widetilde{P}_b^T(T) = P_b(T^{(\ell)}, P) + \left. \frac{\partial P_b(T, P)}{\partial T} \right|_{T=T^{(\ell)}} \left(T - T^{(\ell)} \right). \quad (144)$$

This approximation is locally tight at $T^{(\ell)}$, since it matches both the function value and the first-order derivative of the original BER term. Replacing $P_b(T, P)$ in P5-T with $\widetilde{P}_b^T(T)$ yields a tractable surrogate problem for updating $\mathbf{T}$.

Given the updated transmission-time allocation $\mathbf{T}^{(n+1)}$, the power-allocation subproblem is defined as

$$\text{(P5-P)} \quad \min_{\mathbf{P}} \quad \frac{1}{\sum_{i=1}^{m} r_i} \sum_{i=1}^{m} \sum_{j=1}^{D_i} r_i \alpha_{i,j} \beta_{i,j} \zeta_{i,j} P_{b,i,j}(T_{i,j}^{(n+1)}, P_{i,j}) \quad (145a)$$

$$\text{s.t.} \quad \sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} T_{i,j}^{(n+1)} P_{i,j} \le \bar{P} \sum_{i=1}^{m} \sum_{j=1}^{D_i} c_{i,j} N_{i,j} T_{i,j}^{(n+1)}, \quad (145b)$$

$$P_{i,j} \ge 0, \quad \forall i, j. \quad (145c)$$

Similarly, with T fixed, the first-order approximation of the BER term at the current SCA point $P^{(\ell)}$ is

$$\widetilde{P}_b^P(P) = P_b(T, P^{(\ell)}) + \left. \frac{\partial P_b(T, P)}{\partial P} \right|_{P=P^{(\ell)}} \left(P - P^{(\ell)} \right). \quad (146)$$

Replacing $P_b(T, P)$ in P5-P with $\widetilde{P}_b^P(P)$ gives the corresponding surrogate problem for updating $\mathbf{P}$. The updated solution is then used as the expansion point in the next iteration.

The first-order approximations used in the SCA steps are locally tight, i.e., they match both the function value and the first-order derivative of the original BER expression at the current expansion point. Since P5 is nonconvex, the obtained solution is not guaranteed to be globally optimal. Instead, the AO-SCA procedure provides an efficient suboptimal feasible solution for the digital Rayleigh case.

K Experimental details

Table 1 summarizes the model–dataset pairs used for the seven UE requests. The accuracy denotes the error-free inference accuracy before wireless transmission distortion. The parameter number is measured in millions and is used to determine the transmission workload of each UE.

Algorithm 2 AO–SCA Algorithm for Digital Transmission over Rayleigh Fading

Require: Layer coefficients $\{\alpha_{i,j}, \beta_{i,j}, \zeta_{i,j}\}$, modulation parameters $\{K_{i,j}, \eta_{i,j}\}$, channel statistics $\{\rho_i\}$, system constraints, maximum number of iterations $I_{\max}$, and convergence tolerance ϵ.
Ensure: Feasible allocation $\{\mathbf{T}, \mathbf{P}\}$.

```
Initialize a feasible solution {T^(0), P^(0)}.
Compute the initial objective value F^(0) of (P5).
Set n = 0.
while n < I_max do
    Given P^(n), solve (P5-T) by SCA and obtain T^(n+1).
    Given T^(n+1), solve (P5-P) by SCA and obtain P^(n+1).
    Compute the objective value F^(n+1) of (P5).
    if |F^(n+1) − F^(n)| / F^(n) ≤ ε then
        break
    end if
    n ← n + 1.
end while
return {T^(n+1), P^(n+1)}.
```

Table 1: Model–dataset pairs used in the experiments.

UE	Model	Dataset	Params (M)	Acc. (%)
1	ResNet-18	CIFAR-10	11.16	93.00
2	ResNet-18	CIFAR-100	11.21	68.37
3	ResNet-18	ImageNet	11.68	70.02
4	VGG-16	ImageNet	138.34	71.25
5	ShuffleNetV2	CIFAR-100	0.44	67.81
6	MobileNetV2	CIFAR-10	2.22	94.05
7	MobileNetV2	CIFAR-100	2.33	74.31